TheLook Sales Performance Analysis

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Business Context

Company Overview

TheLook is a fashion e-commerce retailer developed by Looker team. It sells clothing and accessories online, manages product inventory across distribution centers, and serves customers across multiple regions.

Objectives

The company aims to optimize inventory management and strengthen customer loyalty by analyzing monthly inventory growth and monthly retention cohorts,

SQL References

https://console.cloud.google.com/bigquery?ws=!1m7!1m6!12m5!1m3!1srevou-sql-class-479302!2sus-central1!3s3b58e050-4834-4964-a3cc-d5de725c096d!2e1

Disclaimer

- This analysis is based on a public dataset
- *TheLook* is a fictional company created for analytical purposes.
- Any insights or recommendations are based on this dataset and do not reflect real-world data.

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1. Business problem

1. What is the Monthly Growth of Inventory per Category?
2. What is the Monthly Retention Cohorts?

S	Specific – target a specific area for improvement	Inventory	Retention
M	Measurable – quantify or show an indicator of progress Applied to the ideal state:	Monthly Growth of Inventory per Category	Monthly Retention Cohorts
A	Achievable – they need to be agreed, to be attainable and able to be implemented	Result is already achieved	Result is already achieved
R	Realistic – states what results can realistically be achieved, given available resources	Realistic from past data in year 2022	Realistic from past data in year 2022
T	Time-bound - there need to be deadlines, but are they reasonable?	Monthly data in year 2022	Monthly data in year 2022

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2. Important Key Column For Each Table

Table	Key Column
distribution_centers	id name
events	id user_id sequence_number session_id created_at event_type
inventory_items	id product_id created_at sold_at product_category product_distribution_center_id

Table	Key Column
order_items	id order_id user_id product_id inventory_item_id status created_at
orders	order_id user_id Status created_at
products	id category distribution_center_id
users	id created_at

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3. Use Case of the Data to Support Business Objectives

Business problem	Use Case
Monthly Growth of Inventory per Category	• Supply Chain Optimization • The company needs to monitor monthly inventory growth by product category to ensure stock levels are aligned with demand and to prevent overstock or shortages. • Identify fast-growing categories that require more stock allocation. • Detect declining categories to avoid overstock and reduce holding costs. • Support supply chain planning and distribution center efficiency
Monthly Retention Cohorts	• Customer Relationship Management. • The company must understand how many customers return after their first purchase to measure loyalty and improve marketing and engagement strategies. • Measure how many customers return after their first purchase. • Spot patterns of churn and loyalty across different months • Inform marketing campaigns and customer engagement strategies.

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4. Use the ERD To Know the Source of Each Column in the Final Table per Question.

Final Table for Monthly Growth of Inventory

Column	Source
Product Category	fsda-sql-01.TheLook_Ecommerce.inventory_items.product_category
Month	fsda-sql-01.TheLook_Ecommerce.inventory_items.created_at
Growth Percentage	fsda-sql-01.TheLook_Ecommerce.inventory_items.id

Final Table for Monthly Retention Cohorts

Column	Source
Month first purchase	fsda-sql-01.TheLook_Ecommerce.orders.created_at
Month return	fsda-sql-01.TheLook_Ecommerce.orders.created_at
Number of users at first purchase (cohort_size)	fsda-sql-01.TheLook_Ecommerce.orders.user_id
Number of users return (returning_users)	fsda-sql-01.TheLook_Ecommerce.orders.user_id
Retention Rate	fsda-sql-01.TheLook_Ecommerce.orders.created_at fsda-sql-01.TheLook_Ecommerce.orders.user_id

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5. Monthly Growth Of Inventory By Product Categories

Schema | Details | Preview | Table Explorer | Preview

Filter Enter property name or value

Field name	Type	Mode	D
Product_Category	STRING	NULLABLE	-
Month	STRING	NULLABLE	-
Growth_Percentage	FLOAT	NULLABLE	-

Link table :

https://console.cloud.google.com/bigquery?ws=!1m5!1m4!4m3!1srevou-sql-class-479302!2sTheLook!3sMonthly_Inventory_Growth

Row	Product_Category	Month	Growth_Percenta...
1	Accessories	2022-12	5.44
2	Accessories	2022-11	2.15
3	Accessories	2022-10	8.75
4	Accessories	2022-09	-3.11
5	Accessories	2022-08	12.35
6	Accessories	2022-07	2.48
7	Accessories	2022-06	-2.02
8	Accessories	2022-05	10.09
9	Accessories	2022-04	-7.29
10	Accessories	2022-03	18.02
11	Accessories	2022-02	-15.15
12	Accessories	2022-01	null
13	Active	2022-12	2.4
14	Active	2022-11	1.34
15	Active	2022-10	8.45
16	Active	2022-09	-2.2
17	Active	2022-08	7.05
18	Active	2022-07	13.15
19	Active	2022-06	-6.03
20	Active	2022-05	4.94
21	Active	2022-04	-5.4
22	Active	2022-03	16.89
23	Active	2022-02	-10.81
24	Active	2022-01	null

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5. Monthly Growth Of Inventory By Product Categories

Insights

- **Volatility across categories:**
  Every category shows swings (positive and negative) month to month.
  This suggests inventory management is reactive rather than planned—stock is being replenished or depleted in bursts instead of smooth cycles.
- **Seasonal restocking patterns:**
  Peaks in **certain months** → seasonal demand (new collections, holiday season, promotions).
  Dips in **certain months** → stockouts, clearance, or supply chain delays.
- **Operational inefficiency:**
  The fact that volatility is consistent across categories means it's not just product-specific.
  It points to **system-wide issues**: forecasting, supplier coordination, or warehouse planning.
- **Risk of mismatch:**
  Sharp negative growth months could mean **lost sales opportunities** (stockouts)
  Sharp positive growth months could mean **overstock risk** (tying up capital in inventory).

```sql
WITH monthly_inventory AS (
    -- Hitung jumlah item per kategori per bulan di tahun 2022
    SELECT
        i.product_category,
        FORMAT_TIMESTAMP('%Y-%m', i.created_at) AS month,
        COUNT(i.id) AS inventory_count
    FROM `fsda-sql-01.TheLook_Ecommerce.inventory_items` i
    -- contoh join ke tabel products untuk memenuhi syarat "JOIN"
    JOIN `fsda-sql-01.TheLook_Ecommerce.products` p
      ON i.product_id = p.id
    WHERE EXTRACT(YEAR FROM i.created_at) = 2022
    GROUP BY product_category, month
),

growth_calc AS (
    -- Hitung growth percentage dengan window function
    SELECT
        product_category,
        month,
        inventory_count,
        LAG(inventory_count) OVER (
            PARTITION BY product_category
            ORDER BY month
        ) AS prev_inventory_count
    FROM monthly_inventory
)

SELECT
    product_category AS Product_Category,
    month AS Month,
    CASE
        WHEN prev_inventory_count IS NULL THEN NULL
        ELSE ROUND(
            (inventory_count - prev_inventory_count) * 100.0 / prev_inventory_count, 2
        )
    END AS Growth_Percentage
FROM growth_calc
ORDER BY product_category, month DESC;
```

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6. Monthly Retention Cohorts

Schema | Details | Preview | Table Explorer | Preview

Filter Enter property name or value

Field name	Type	Mode
Month_first_purchase	STRING	NULLABLE
Month_return	STRING	NULLABLE
cohort_size	INTEGER	NULLABLE
returning_users	INTEGER	NULLABLE
Retention_Rate	FLOAT	NULLABLE

Link table :

https://console.cloud.google.com/bigquery?ws=!1m5!1m4!4m3!1srevou-sql-class-479302!2sTheLook!3sRetention

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Month_first_purchase	Month_return	cohort_size	returning_users	Retention_Rate
1	2022-01	2022-01	750	750	100.0
2	2022-01	2022-02	750	8	1.07
3	2022-01	2022-03	750	9	1.2
4	2022-01	2022-04	750	9	1.2
5	2022-01	2022-05	750	11	1.47
6	2022-01	2022-06	750	4	0.53
7	2022-01	2022-07	750	5	0.67
8	2022-01	2022-08	750	6	0.8
9	2022-01	2022-09	750	8	1.07
10	2022-01	2022-10	750	12	1.6
11	2022-01	2022-11	750	6	0.8
12	2022-01	2022-12	750	6	0.8
13	2022-02	2022-02	652	652	100.0
14	2022-02	2022-03	652	11	1.69
15	2022-02	2022-04	652	6	0.92
16	2022-02	2022-05	652	4	0.61
17	2022-02	2022-06	652	6	0.92
18	2022-02	2022-07	652	7	1.07
19	2022-02	2022-08	652	5	0.77
20	2022-02	2022-09	652	11	1.69
21	2022-02	2022-10	652	9	1.38
22	2022-02	2022-11	652	9	1.38
23	2022-02	2022-12	652	7	1.07

6. Monthly Retention Cohorts

Insights

- **Sharp drop-off after the first month**:
  Only about 1-2% returned in the following months.
  This means **most users purchased only once**.

- **Extremely low retention**:
  After the initial month, **retention rate** consistently stays **< 3%**.
  This indicates **weak customer loyalty** or one-time purchase behavior.

- **Conclusion**:
  Need for **stronger customer retention strategies** (loyalty programs, repeat purchase incentives, personalized recommendations) to reduce the steep drop-off after the first purchase.

```sql
WITH first_purchase AS (
    SELECT
        o.user_id,
        FORMAT_TIMESTAMP('%Y-%m', MIN(o.created_at)) AS cohort_month
    FROM `fsda-sql-01.TheLook_Ecommerce.orders` o
    WHERE o.status = 'Complete'
      AND EXTRACT(YEAR FROM o.created_at) = 2022
    GROUP BY o.user_id
),

user_orders AS (
    SELECT
        o.user_id,
        FORMAT_TIMESTAMP('%Y-%m', o.created_at) AS order_month
    FROM `fsda-sql-01.TheLook_Ecommerce.orders` o
    WHERE o.status = 'Complete'
      AND EXTRACT(YEAR FROM o.created_at) = 2022
),

cohort_activity AS (
    SELECT
        f.cohort_month,
        u.order_month,
        u.user_id
    FROM first_purchase f
    JOIN user_orders u
      ON f.user_id = u.user_id
    WHERE u.order_month >= f.cohort_month
),

cohort_size AS (
    SELECT
        cohort_month,
        COUNT(DISTINCT user_id) AS cohort_size
    FROM cohort_activity
    WHERE cohort_month = order_month
    GROUP BY cohort_month
),

retention_calc AS (
    SELECT
        cohort_month AS Month_first_purchase,
        order_month AS Month_return,
        COUNT(DISTINCT user_id) AS returning_users
    FROM cohort_activity
    GROUP BY cohort_month, order_month
)

SELECT
    r.Month_first_purchase,
    r.Month_return,
    c.cohort_size,
    r.returning_users,
    ROUND(r.returning_users * 100.0 / c.cohort_size, 2) AS Retention_Rate
FROM retention_calc r
JOIN cohort_size c
  ON r.Month_first_purchase = c.cohort_month
ORDER BY r.Month_first_purchase, r.Month_return;
```

Summary of Key Findings

1. **High Volatile Monthly Growth of Inventory across all categories:**
   Every category shows swings (positive and negative inventory growth) month to month.

2. **Extremely low retention rate (< 3%)**

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Recommendation

Insights / Observation	Business Impact	Isolation	Prioritization	Recommendation
High Volatile Monthly Growth of Inventory across all categories: Every category shows swings (positive and negative inventory growth) month to month.	Risk of mismatch: Sharp negative growth months could mean lost sales opportunities (stockouts) Sharp positive growth months could mean overstock risk (tying up capital in inventory). Volatility is systemic, not isolated. The business needs to move from reactive inventory management to proactive, forecast-driven planning.	Forecasting	High High impact – High effort – Many data points	1. Better demand forecasting: Smooth out replenishment cycles by predicting demand more accurately, instead of reacting after stockouts. 2. Align procurement with seasonality: Plan restocks ahead of known demand peaks (holiday, fashion cycles). 3. Inventory buffers: Maintain safety stock in volatile categories to reduce risk of sudden dips. 4. Category prioritization: Focus on stabilizing high-value categories first (e.g., Suits, Accessories) where volatility has bigger financial impact.
Extremely low retention rate (< 3%)	Increase retention rate can boost revenue	Retention rate	High High impact – High effort – Many data points	Need for stronger customer retention strategies (loyalty programs, repeat purchase incentives, personalized recommendations)

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THANK YOU

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